Test Design techniques Advanced

Part I: Equivalence classes

Test scenario description	Expected result
State field:	
Positive class: User enters two capital letters, which is an abbreviation for states. Test data: LA.	The state is determined and accepted.
Positive class: User enters two lowercase letters, that are abbreviations for states. Test data: an.	The state is determined and accepted.
Negative class: Users enter two capital letters that are not abbreviations for states. Test data: OG.	Error message: no such state exists.
Negative class: User enters two lowercase letters that are not abbreviations for states. Test data: zx.	Error message: no such state exists.
Negative class: User enters one character. Test data: q.	Error message: the standard Post Office abbreviation for the states consists of two capital letters.
Negative class: Users enter more than two letters. Test data: SDFGH.	Error message: the standard Post Office abbreviation for the states consists of two capital letters.

Test scenario description	Expected result
Last Name field:	
Positive class: User enters 8 characters. Test data: Khomenko.	The last Name is accepted.
Negative class: Users stay in this field empty.	Error message: The last Name field should have one through fifteen characters.
Negative class: User enters 26 characters. Test data: ghjkkjdnyue44bnsdhj45nmdk.	Error message: The last Name field should have one to fifteen characters.
Negative class: User enters 12 characters with two unacceptable symbols. Test data: Kolyadenko$%	Error message: The last name field shouldn’t contain symbols.

Test scenario description	Expected result
User ID field:	
Positive class: User enters 2 not alphabetic symbols and 6 letters. Test data: SA12SAJA.	User ID is accepted.
Negative class: User enters 6 characters. Test data: hklm34	Error message: User ID field should enter by 8 characters at least two of which are not alphabetic (numeric, special, nonprinting).
Negative class: User enters 10 characters. Test data: Koshka23#%.	Error message: User ID field should enter by 8 characters at least two of which are not alphabetic (numeric, special, nonprinting).
Negative class: User enters 1 not alphabetic symbol and 7 letters. Test data: SA12SAJA.	Error message: User ID field should enter by 8 characters at least two of which are not alphabetic (numeric, special, nonprinting).

Test scenario description	Expected result
Student ID field:	
Positive class: User enters eight characters. The first two represent the student's home campus while the last six are unique six-digit numbers. Test data: LC235698.	Student ID is accepted.
Negative class: User enters 6 characters. Test data: hklm34.	Error message: Student ID field should consist of eight characters. The first two represent the student's home campus while the last six are unique six-digit numbers.
Negative class: User enters 10 characters. Test data: WN90236789	Error message: Student ID field should consist of eight characters. The first two represent the student's home campus while the last six are unique six-digit numbers.
Negative class: User enters eight characters. The first two represent the false student's home campus while the last six are unique six-digit numbers. Test data: LL235698.	Error message: Enter the right student's home campus.
Negative class: User enters eight characters. The first two represent the student's home campus while the last six have one symbol. Test data: LC23%698.	Error message: Enter the right unique number.

Part II: Boundary values

Test scenario description	Expected result
ZIP Code	
Positive scenario: ZIP Code 01000.	ZIP Code is accepted.
Positive scenario: ZIP Code 99000.	ZIP Code is accepted.
Negative scenario: ZIP Code 00099.	Error message: This zip code does not exist.
Negative scenario: ZIP Code 99001.	Error message: This zip code does not exist.

Test scenario description	Expected result
Last Name	
Positive scenario: User enters one character in the Last Name field. Test data: W.	Last Name is accepted.
Positive scenario: User enters 15 characters in the Last Name field. Test data: Worthjkllalonak.	Last Name is accepted.
Negative scenario: Last Name field is empty.	Error message: Last Name field shouldn’t be empty.
Negative scenario: User enters 16 characters in the Last Name field. Test data: Chudovsky-Volkov.	Error message: Last Name is too large.

Test scenario description	Expected result
User ID	
Positive scenario: Users enter eight characters at least two of which are not alphabetic (numeric, special, nonprinting) in the User ID field. Test data: GRUSHA12.	User ID is accepted.
Negative scenario: Users enter seven characters at least two of which are not alphabetic (numeric, special, nonprinting) in the User ID field. Test data: TIKA2#.	Error message: User ID field should consist of eight characters at least two of which are not alphabetic (numeric, special, nonprinting).
Negative scenario: Users enter nine characters at least two of which are not alphabetic (numeric, special, nonprinting) in the User ID field. Test data: Glushko10	Error message: User ID should consist of eight characters at least two of which are not alphabetic (numeric, special, nonprinting).

Test scenario description	Expected result
Course ID	
Positive scenario: User enters three alpha characters and six numbers. Test data: FIT120101.	Course ID is accepted.
Negative scenario: two alpha characters and six numbers. Test data: FI120101.	Error message: Course ID should consist of three alpha characters and six numbers.
Negative scenario: Four alpha characters and six numbers. Test data: FIKT120101.	Error message: Course ID should consist of three alpha characters and six numbers.